

A Packing of 11 Four-Spaces in $\mathbb{R}^9$

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October 9, 2026

Abstract

We give 11 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.444131. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.423926, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 11 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.444131 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 11$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.423926, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 11/10 = 2.444444$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N11.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
152758235856724069934520047167227800081743720903225074753281794139290040
289353868021706456251494198531866695744038727385145336648667900547075719
55850348990059985399394204550610924404475987183
```

denominator:

```
624999999998896042946627679088042388340974226239563490421196707837717924
423103065616746602007134598196955570830612295566884295041732295393886329
1520576810672456286553881372938395204654784559
```

B Coordinates

11 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N11.txt`.

```
-36788810183 519147167872 -344546461459 -4788310758263927396362 -311517749804 136394312760 29663648206213814336952 231753654409 141762390116 30257118334
-303387984400 234651056552 216030016046 -4871118023350721801485 -644378274312 -187161752446 -40591905798589358944 374353938791 249717081883 -414015619044
-200511657434 -231898833625 658508023715 -603519276338938009974 461731549586 -54768304206 57910026953 102540620375 386615411848 -366250000291 -557185037824
-372027686850 500667065397 231605582635 4522766428014252474870 207862645258 -402824829434 -64030954199136875281688 371007829383 -326134087247 14746985341
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-432087841204 212275649584 24091819426 -1701500545491568116566 -185199208394 -106873828222 4263756512063341531775 151301731065 189241834367 -591595749579
60743915455 553716924074 567655082324 169322219052 322447694101 538177069361 623294611267 -149163468582160453284249 319621285498 -130253335685 79128742184
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-248946440541 -394536114770 686000328966 -1246885512138433658012 348487570309 160008564985 6840135110389432587158 362749644079 -496865889739 -139091249154
344491161402 -62636436446 93725537546 -668786141991-10538585091 494001176570 171222050663 -21958436072426180956250 64499926763 78239551762 218411399587
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```